

PLAIN-ENGLISH PROJECT BRIEF

EcoBrief

Your AI-Powered Daily News Briefing, Explained Simply

Everything you need to know about this project — what it does, how it works, why it costs nothing to run, and where it's going — written for someone who has never written a line of code in their life.

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May 2026

— SECTION 01

What Is EcoBrief?

"Imagine waking up every morning and instead of doom-scrolling through 200 news articles, your phone simply plays you a crisp 2-minute audio summary of everything important that happened overnight — written and narrated entirely by AI, for free."

That's EcoBrief. It is a piece of software I built that automatically reads the internet for you, uses artificial intelligence to squeeze the most important parts into a short summary, and then reads that summary back to you in a natural-sounding voice — all without any human touching it in between.

Think of it like a very smart, very fast personal assistant who:

1**Reads thousands of news articles every day**

It connects to live news feeds across the internet — the same raw data that apps like BBC News or Reuters publish publicly — and pulls all of it in automatically.

2**Understands and summarises them using AI**

A cutting-edge AI language model (similar to the technology behind ChatGPT, but running privately on our own servers) reads each article and writes a clean, concise summary.

3

Converts the summary into spoken audio

Amazon's text-to-speech service turns that written summary into a high-quality MP3 audio file — the kind of voice quality you'd hear on a professional podcast.

4

Shows everything in a clean dashboard

You open a web page, see today's article summaries listed out, click one, and hit play. That's it. Read or listen, your choice.

5

Shuts itself down to cost nothing

The moment it's done processing, every piece of heavy cloud computing it used turns itself off automatically. The bill at the end of the month? Essentially zero.

**The one-sentence version**

EcoBrief is a robot that reads the news, summarises it with AI, speaks it aloud, and costs almost nothing to run because it's clever enough to only use computing power when it actually needs to.

— SECTION 02

The Problem It Solves

There are three big problems with the way most people consume news today, and EcoBrief directly tackles all three:

**PROBLEM #1**

Information Overload

The average person is exposed to the equivalent of 174 newspapers of information every single day. Nobody has time to read it all, so most people read nothing properly.

**PROBLEM #2**

Expensive AI Tools

Running AI to summarise content is normally very expensive — companies pay thousands per month in cloud computing bills to do what EcoBrief does for free.

**PROBLEM #3**

Screen Fatigue

People are tired of staring at screens. Being able to press play and listen while you make breakfast or commute is genuinely valuable.

**PROBLEM #4**

Data Privacy

Most AI news apps send your reading habits to third-party companies. EcoBrief runs the AI privately on its own server — your data stays yours.

The "Zero Cost" Superpower 🚀

The single most impressive thing about EcoBrief isn't what it does — it's how cheaply it does it. Running an AI system like this normally costs hundreds to thousands of dollars per month. EcoBrief's entire architecture was designed from the ground up to keep that bill at effectively **\$0**.

Here's the analogy: most AI services leave their servers running 24/7 like leaving every light in your house on all day and night. EcoBrief's servers only switch on when someone actually needs them — and switch off the second they're done. Like a smart light switch on steroids.

\$0

Monthly cloud bill (target)

3B

Parameter AI model running on it

2 GB

RAM it runs within

100%

Auto shutdown after each job

— SECTION 03

The Four Parts of EcoBrief

EcoBrief is made up of four separate pieces of software, each with one very specific job.

Think of them like departments in a company — they each do their thing, then hand the work off to the next department.

Part 1 — The News Collector (Ingestion Service)

What it does in plain English: This is the robot librarian. It goes out onto the internet, visits news websites, downloads all the article text, cleans it up (removes junk, ads, duplicate content), and saves it into a storage folder in the cloud.

What it's built with: Java — a programming language well known for being extremely reliable and fast when handling large amounts of data. Think of Java as the workhorse language used in most banking software.

Where it saves the data: Amazon S3. This is basically a giant, virtually unlimited hard drive in the cloud. It costs fractions of a penny per gigabyte to store things there. Your data is always safe and retrievable instantly.



Simple analogy

Imagine hiring a very fast reader to go through every newspaper published today, rip out the useful articles, bin the adverts, and stack the articles neatly in a folder. That folder lives in a fireproof room (Amazon S3) nobody can break into.



Part 2 — The Traffic Controller (Orchestrator)

What it does in plain English: This is the manager in the middle. When you click a button on the website asking for a news briefing, this manager receives that request, checks whether it's already been done before (to avoid wasted work), and if not — it "wakes up" a powerful computer in Amazon's data centre specifically to do the AI processing.

What it's built with: Python and FastAPI. Python is considered the world's most readable programming language — the one most scientists and AI researchers use. FastAPI makes it possible to build a super fast web service with it.



Simple analogy

Think of this like a restaurant manager. When a table orders food, the manager checks if the kitchen needs to open up (wake up the AI computer), fires up the grill, and gives the kitchen the order. When the food is ready and delivered, the manager turns the grill off again so you're not paying gas bills for an empty kitchen.



Part 3 — The AI Brain (Inference Node)

What it does in plain English: This is the actual AI doing the smart work. A computer in Amazon's cloud wakes up, loads an AI model into its memory (specifically a 3-billion-parameter language model — think of "parameters" as the number of learned facts the AI has stored), reads each news article, and writes a clear summary of it. It then sends that summary to Amazon's voice service to generate an audio file.

What it's built with: llama.cpp (a highly optimised AI engine) running on a Llama language model. This is the same family of AI that Meta (Facebook's parent company) released openly to the world. It's genuinely impressive technology — capable of understanding context, nuance, and complex sentences — running on very modest hardware.



Simple analogy

This is like hiring a brilliant journalist who has read millions of articles and can produce a tight, accurate 200-word summary of any story in under a minute. Except this journalist never sleeps, never gets tired, and charges nothing. And when the job is done, they disappear until needed again.



Part 4 — The Dashboard (Frontend / User Interface)

What it does in plain English: This is the part you actually see and touch. A clean, dark-themed web page that shows you all the processed news articles in a list. Each one shows when it was processed, its size, and whether the AI has finished working on it. You click one, read the AI's summary, and click the play button to hear it spoken aloud.

What it's built with: React — the same technology used to build Facebook, Instagram, Netflix, and Airbnb's web interfaces. It's fast, smooth, and handles live updates well (so the status of an article updates in real-time without you needing to refresh the page).



Simple analogy

Think of this like the Spotify app, but instead of music, it's playing you AI-generated audio news summaries. The interface is minimal and elegant — the only thing you need to do is scroll and press play.

— SECTION 04

How It All Flows Together

Here is the complete journey, from a news article existing on the internet to you hearing a summary of it in your earphones. This entire process happens automatically, without any human intervention.

A**The internet publishes news**

News outlets like Reuters, BBC, TechCrunch, etc. publish articles via RSS feeds — a standard format for broadcasting new content. EcoBrief connects to these feeds on a schedule (e.g. every few hours).

B**The Collector grabs and stores it**

The Java-based news collector downloads all new articles, strips out HTML formatting and advertisements, leaving just the clean article text. This gets saved to Amazon S3 with a timestamp and unique ID.

C**You click "Process" on the dashboard**

On the web interface, you see a list of collected articles. You click a button to trigger the AI summarisation. The Orchestrator (Traffic Controller) receives this signal.

D**A cloud computer wakes up just for this job**

The Orchestrator sends a command to Amazon Web Services: "Start a computer for me." Within 60–90 seconds, a fresh ARM-based server in Amazon's data centre boots up, downloads the article text from S3, and starts the AI model.

E**The AI reads and summarises**

The 3-billion-parameter language model reads the full article text and produces a concise, accurate summary — typically 150–300 words. This summary is structured as a proper news briefing: headline, key facts, context.

F

Amazon's voice service reads it aloud

The summary text is sent to AWS Polly — Amazon's text-to-speech service which produces studio-quality audio. The resulting MP3 file gets uploaded back to Amazon S3 for storage and retrieval.

G

The cloud computer shuts itself off

The moment the audio file is saved, the server terminates its own existence. It sends a self-destruct command to Amazon. From this moment forward, Amazon stops charging anything for it. The entire compute cost for one article typically amounts to a fraction of one cent.

H

You read and listen on the dashboard

Your web browser polls for updates. The article card changes from "Processing" to "Ready." You click it, read the summary on screen, or hit the play button and listen to the MP3 audio. Done.

— SECTION 05

What Is AWS and Why Are We Using It?

AWS stands for **Amazon Web Services**. It is essentially Amazon's cloud computing division — they rent out computing power, storage, and hundreds of other services over the internet to developers, startups, and corporations worldwide.

Netflix streams your movies from AWS. Airbnb runs its booking platform on AWS. NASA uses AWS for space data. It is the backbone of the modern internet.

Why EcoBrief uses AWS specifically

FREE**KEY REASON**

AWS Free Tier

New AWS accounts get 12 months of free access to a bundle of services — including 750 hours of computing time and 5 GB of storage. EcoBrief is designed to stay entirely within these free limits.

**RELIABILITY**

Global Infrastructure

Amazon has data centres on every continent. If one data centre has an issue, your data and your service automatically moves to another. This level of reliability would cost millions to build yourself.

**INTEGRATION**

AWS Polly (Voice AI)

Amazon's own text-to-speech service produces remarkably natural-sounding

**STORAGE**

Amazon S3

Think of S3 as a giant, infinitely expandable USB stick in the cloud. Files stored there

speech. It is deeply integrated with the rest of AWS, making it trivially easy to use from within EcoBrief's code.

are virtually indestructible (Amazon replicates them across multiple data centres automatically) and accessible from anywhere.



The "zero cost" strategy in detail

The way most cloud bills become expensive is when you leave servers running 24/7. EcoBrief only ever turns on a full AI server when a briefing needs to be generated — and that process takes roughly 2–5 minutes. At AWS Free Tier pricing, that's a rounding error. The rest of the time, the only thing running is a tiny "manager" server (the Orchestrator), which is so lightweight it fits comfortably within the AWS Free Tier's 750 monthly free hours.

The specific AWS services being used

EC2 (Elastic Compute Cloud)

Virtual computers in Amazon's data centre. EcoBrief uses **t2.micro** (a tiny, free-tier eligible computer) for the always-on manager, and **t4g.small** (a slightly beefier ARM-chip computer) specifically for the brief burst of AI processing. The latter terminates itself within minutes of each job completing.

S3 (Simple Storage Service)

Two separate storage buckets: one for raw collected news articles (the "inbox"), and one for finished audio MP3 files (the "outbox"). Files stored here persist indefinitely and can be accessed at any time.

AWS Polly

Amazon's speech synthesis service. You send it text, it sends you back an MP3. The voice quality is dramatically better than older text-to-speech systems — it handles pauses, emphasis, and natural cadence well. Completely free for the first 5 million characters per month under the Free Tier.

IAM (Identity and Access Management)

The security layer. This is how AWS knows "this particular piece of software is allowed to start and stop computers, but not allowed to access billing information" etc. EcoBrief uses an IAM user called **mesh** with the exact minimum permissions needed to do its job — nothing more.

— SECTION 06

What the User Actually Sees

EcoBrief's interface is deliberately minimal. There are no confusing menus, no account creation, no complicated settings. Here's a walk-through of exactly what you see when you open it:

The Header

A dark navigation bar at the top with the EcoBrief logo and a small pulsing green dot indicating the system is live and running. That's it. Clean.

The Article Grid

Below the header, you see a grid of cards — one for each news article that has been collected. Each card shows:



Article filename and size

A human-readable name for the article and how large the raw file was (in kilobytes). Larger files generally mean longer articles.



Status badge

Either **Ready** (AI has processed it, audio is available) or **Pending** (collected but not yet summarised). This updates live without you refreshing the page.



Date and time

When the article was collected. Always shown so you know how fresh the information is.

► Trigger Button

A button on each card that, when clicked, tells the system to process that article and generate an audio briefing. A loading spinner appears while the AI is working.

The Briefing Viewer

Once you click into a "Ready" article, you see a full-screen panel with:



The written summary

The AI's clean, readable summary of the article. Typically 150–300 words. Written in plain English with no jargon. Structured with a headline, the core facts, and any important context.



The audio player

A standard audio player bar — press play, adjust volume, scrub through. The audio is the AI summary spoken aloud in Amazon Polly's voice. Typically 60–90 seconds long per article.



The design language

Everything uses a deep dark navy background (#080e1a), glassmorphic card borders that glow subtly on hover, clean white typography, and smooth animations on state changes. It feels premium despite being built on zero budget.

— SECTION 07

The Technology Stack — In Plain English

A "technology stack" is the list of tools and languages used to build a piece of software.

Here's every single tool in EcoBrief and what it actually does, explained without any jargon:

Java

The programming language used to build the news collector. Java is one of the oldest and most reliable programming languages in the world. It runs on billions of devices — from Android phones to Wall Street trading systems. It was chosen here because it's exceptionally good at handling large data processing jobs quickly.

Python

The programming language used for the orchestrator (traffic controller) and AI inference code. Python is beloved by AI researchers worldwide because it reads almost like English, has a massive library of AI tools, and is fast to write. It's the language OpenAI, Google DeepMind, and most AI labs use.

FastAPI

A Python framework that makes it easy to build fast web APIs (programming interfaces that let different software talk to each other). Think of an API as a waiter in a restaurant — it takes your order (request) to the kitchen (server) and brings back the food (response).

Llama-3.2-3B (the AI model)

This is the actual AI brain. A "3B parameter" language model means an AI that has stored 3 billion pieces of learned information from training on vast amounts of text. Meta (Facebook) released this model publicly, and we run it in a "quantized" format — meaning

it's been compressed to use less memory while retaining most of its intelligence. It's like compressing a huge video file without losing much visible quality.

llama.cpp / Ray Serve

The engine that runs the AI model. llama.cpp is an extremely optimised piece of software that can run large language models on very modest hardware — specifically, it was engineered to run on regular laptops without a specialised GPU graphics card. Ray Serve wraps around it to handle multiple requests efficiently.

React + Vite

The technology powering the web interface. React is a JavaScript library created by Facebook used to build fast, interactive web pages. Vite is a development tool that makes building React apps dramatically faster. Together they're the standard choice for modern web applications — Netflix, Airbnb, and Instagram all use React.

Tailwind CSS

The styling system for the web interface. CSS is what controls how a webpage looks — fonts, colours, layout, spacing. Tailwind makes it faster to write good-looking designs by providing thousands of pre-defined style building blocks.

Docker

A technology that packages each piece of software into a self-contained "container." Think of Docker containers like shipping containers — the same container (software package) works identically on any ship (computer), whether it's your laptop or Amazon's servers in Virginia. This eliminates "it works on my computer but not yours" problems entirely.

Boto3

Python's official library for talking to AWS. It's the code that says "hey Amazon, start a new server for me" and "hey Amazon, turn this server off now." Without Boto3, the zero-cost scale-to-zero strategy would be impossible to automate.

SECTION 08

Deploying on AWS — Making It Live

"Deployment" is the process of taking software from a developer's laptop and putting it on a real internet-connected server so that anyone in the world can use it. Here's how EcoBrief goes live.

What Deployment Looks Like for EcoBrief

The deployment is designed to be as simple as possible — it's essentially three steps, each triggered by one shell command:

1

Set up your AWS credentials

You create an AWS account, create an IAM security user (called "mesh" in our case), and download a pair of keys — an Access Key and a Secret Key. These act like a username and password that allow our deployment scripts to automatically control AWS on your behalf.

2

Build the AI server image

You start a temporary Amazon computer and run one script:

`build_inference_ami.sh`. This installs all the AI software, creates a "snapshot" of that computer (called an AMI — Amazon Machine Image), and then that computer shuts itself off. This snapshot is what gets cloned every time EcoBrief needs to process an article.

3

Launch the main server and go live

On the main server (a permanently-running t2.micro, which is free), you run `setup.sh`. This installs the orchestrator, installs the frontend, and starts everything running. At the

end it prints your public IP address. Visit that IP in a browser — that's EcoBrief, live on the internet.



What the IAM user "mesh" is

In the AWS account with the ID 339712952304, there is a security user named "mesh." This is the identity that EcoBrief's automated scripts use to control AWS resources. It has exactly the permissions needed: start/stop EC2 instances, read/write S3 buckets, and use AWS Polly. Nothing else. This is best-practice security — minimal access, maximum control.

What Stays Running 24/7 (and costs money)

The t2.micro Orchestrator — FREE

This tiny server runs the web API and serves the frontend. At the AWS Free Tier, you get 750 hours of t2.micro per month completely free — which covers 31 days × 24 hours = 744 hours. The Orchestrator fits within this perfectly.

What Turns On and Off on Demand (costs fractions of a cent)

The t4g.small Inference Node — NEAR FREE

This only runs for 2–5 minutes per article. At approximately \$0.0168 per hour, 5 minutes costs less than \$0.002 — less than a fraction of a cent. Generating 100 briefings per month would cost roughly 20 cents. In practice, within the AWS Free Tier new-account window, even this may be covered.

— SECTION 09

Frequently Asked Questions

Questions a non-technical person is likely to ask, answered honestly:

Is this actually free? What's the catch?

Within the AWS Free Tier (first 12 months of a new AWS account), the costs are effectively zero for reasonable usage. After 12 months, the main server costs roughly \$8–10/month, and each briefing generated costs fractions of a cent. There's no subscription, no per-user fee, and no data being sold. The "catch" is simply that you need an AWS account.

Do I need to be technical to use it?

To use the finished web interface — absolutely not. You open a browser, go to a URL, and click buttons. No technical knowledge required. Setting it up on AWS for the first time requires a developer (about 1–2 hours of work) but once deployed, anyone can use it.

Where does the news come from?

From RSS feeds — a standard publishing format used by thousands of news outlets worldwide. You can configure EcoBrief to follow any news source that has an RSS feed: BBC, Reuters, TechCrunch, The Guardian, Wired, etc. The collector checks for new articles automatically.

Is the AI summary accurate?

Language models like Llama-3.2-3B are very good at summarisation — it's one of the tasks they perform most reliably. However, like any AI, they can occasionally make minor errors in phrasing or miss a nuance. We recommend treating the summaries as a starting

point, not an authoritative source. The original full article is always accessible if you want the complete story.



Is my data private?

Yes. The AI model runs entirely on our own servers — not through OpenAI's API or any other third-party AI service. Nobody else processes your news. The only external service used is AWS Polly for voice generation, which does not store your text after converting it.



Can I use it on my phone?

Yes. The web interface is fully responsive — it adjusts automatically to any screen size. You can open it on any smartphone browser and it works exactly as on desktop, including the audio playback.



How long does it take to generate a briefing?

When you click "Process" on an article, the full cycle takes 3–7 minutes. The largest portion of this is the 60–90 seconds it takes for the inference server to boot up in Amazon's cloud. The actual AI summarisation typically takes under a minute. The audio generation is near-instant.



Can multiple people use it at the same time?

Yes, reading summaries and listening to audio is instant for any number of simultaneous users — those files are already stored. Triggering new AI processing jobs simultaneously is supported but each job spawns its own inference server, so costs would scale linearly. For personal or small-team use, this is not a concern.



What happens if something breaks?

EcoBrief includes a diagnostic script (`verify_finops.py`) that automatically checks whether any servers are still running that shouldn't be. If an inference server gets stuck and doesn't shut itself off, this script detects it and terminates it immediately. There are also manual recovery commands documented in the `RUNBOOK.md` file for any edge cases.



Where is this going — what's next for EcoBrief?

The roadmap includes: personalised topic filtering (only brief me on technology and business news), multi-language support (briefings in French, Spanish, etc.), a mobile app, and the ability to add custom data sources beyond news feeds — such as research papers, legal documents, or earnings reports.

— SECTION 10

Glossary — Every Technical Term Used, Defined

If you've heard a word in this document and weren't sure what it meant, here it is defined in the simplest terms possible:

API (Application Programming Interface)

A structured way for two pieces of software to talk to each other. Like a menu in a restaurant — it defines what you can order (request) and what you'll get back (response).

AWS (Amazon Web Services)

Amazon's cloud computing platform. They rent out servers, storage, and hundreds of other digital services globally.

Cloud

Servers and storage that exist in a data centre somewhere on the internet, rather than on your own physical hardware. "In the cloud" just means "on someone else's computers, accessible via the internet."

Docker / Container

A way to package software and all its dependencies into a self-contained unit that runs identically anywhere. Like a shipping container — same container, any ship.

EC2 (Elastic Compute Cloud)

Amazon's virtual computer rental service. You pay for a computer, Amazon runs it in their data centre, you control it over the internet.

IAM (Identity and Access Management)

AWS's security system for controlling who (or what software) can do what within an AWS account.

Inference

The process of running an AI model to produce an output — in our case, generating a text summary. (Distinct from "training," which is the process of teaching the AI in the first place.)

Language Model (LLM / SLM)

An AI system trained on vast amounts of text that can understand and generate human-like writing. LLM = Large Language Model (billions of parameters). SLM = Small Language Model (still capable but designed to run efficiently on modest hardware).

MP3

A compressed audio file format. The same format used for music files.

Parameters (in AI)

The individual numbers inside an AI model that it learned during training. More parameters generally means a more capable (and more resource-hungry) model. 3 billion parameters is a reasonable balance between quality and cost for EcoBrief's use case.

Quantization

Compressing an AI model to use less memory by reducing the precision of its numbers, while keeping most of its intelligence intact. Like compressing a JPEG image — smaller file, very similar quality.

RSS Feed

A standardised format that news websites use to publish their articles in a machine-readable way. Allows software to automatically check for new articles without scraping web pages manually.

S3 (Simple Storage Service)

Amazon's cloud file storage service. Like Google Drive or Dropbox, but for developers — built for machines to read and write files programmatically.

Scale-to-Zero

The architectural pattern of automatically shutting off all computing resources when they're not needed, so you pay nothing during idle periods.

Server

A computer (usually without a screen, running in a data centre) that serves data to other computers over the internet. When you visit a website, a server somewhere in the world sends your browser those web pages.

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Written by **Sahil Janjua** · May 2026

This document is intended for non-technical audiences. For technical documentation, refer to README.md and RUNBOOK.md.